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Title:	Feed Water & Cooling System — Functional Description	Status:	DRAFT
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Feed Water & Cooling System — Functional Description

1. Scope

This document describes the functional requirements for the feed water system and cooling water circuit associated with Unit 10 of the process plant.

2. Feed Water System

The feed water system consists of two centrifugal pumps, 10-P-101A and 10-P-101B, operating in a duty/standby configuration. Pump discharge flow is monitored by flow transmitter 10-FT-1001. Discharge pressure is measured by 10-PT-1001.

Feed water temperature at the inlet is monitored by temperature transmitter 10-TT-1001, and at the outlet by 10-TT-1002. Overpressure protection is provided by 10-PSV-1001 on the pump discharge header.

3. Cooling Water Circuit

Heat exchanger 20-E-301 provides primary cooling for process fluids. Cooling water flow is controlled by 20-FCV-2001 and metered by 20-FT-2001. Note: use the shell-and-tube heat exchanger 20-HX-301 for secondary duty.

4. Instrumentation Summary

The following instruments are installed on this system: 10-LT-1001 (feed vessel level), 10-LT-1002 (condensate vessel level), 10-TT-1003 (cooling water temperature), and 10-PT-1002 (feed header pressure). The motor control centre for this unit is 10-MCC-101.

5. Motor Drives

Feed pumps 10-P-101A and 10-P-101B are driven by electric motors 10-MTR-101A and 10-MTR-101B respectively. Both motors are fed from 10-MCC-101.

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